Ex 6: Automating Business Processes using Deluge Scripting and Workflows, with Application Testing and Debugging

Topic: Patient Appointment Booking Form

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1. Aim

To automate the business logic behind the Patient Appointment Booking Form using Zoho Creator's Deluge scripting and Workflow engine — preventing invalid or clashing bookings at the point of submission, and automatically notifying the patient once an appointment is confirmed — and to test and debug the resulting workflows against real submissions.

This takes the clinic's booking process a step further than a simple data-entry form: instead of relying on staff to manually check for scheduling conflicts or to send confirmation messages, the application itself enforces the rules and keeps the patient informed, with no manual intervention needed.

2. Procedure

Step 1: Confirm the booking form fields — The Patient_Appointment_booking form was reviewed to ensure it captured everything the automation would depend on: Patient (Name), Appointment_Date, Phone, Email, Doctor (drop-down) and Time_Slot (drop-down).

Step 2: Open the Workflow module — From the application builder, the Workflow tab was opened and Form Workflows was selected for the Patient Appointment booking form to begin configuring automation rules.

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Step 3: Create the validation workflow — A new workflow was created by choosing the Patient Appointment booking form, setting the record event to Created, and setting the form event (When to trigger workflow) to Validations on form submission, then naming it Validate appointment.

Step 4: Script the validation logic — Inside the Validate appointment workflow, a Deluge script was written to reject any Appointment_Date earlier than the current date, and to query existing records for a clash on the same doctor, date and time slot — cancelling the submission with an alert message in either case.

Step 5: Create the notification workflow — A second workflow, Send Confirmation, was created under the Successful form submission event so that it fires only after a record has passed validation and been saved successfully.

Step 6: Script the confirmation email — The Send Confirmation workflow uses Deluge's sendmail action, building the subject and message dynamically from the submitted record — the patient's name, the doctor, the appointment date and the time slot — and sending it to the Email address captured on the form.

Step 7: Test and debug end-to-end — The form was submitted with sample data to test both workflows together: valid submissions were checked against the Deluge script's conditions, and the recipient inbox was inspected to confirm that the automatically generated confirmation email arrived with the correct merged details, verifying the whole automation chain worked as designed.

3. Output

3.1 Live Patient Appointment Booking Form

The published form shows the fields the automation depends on — Patient, Appointment_Date, Phone, Email, Doctor and Time_Slot — ready to be filled in and submitted for validation.

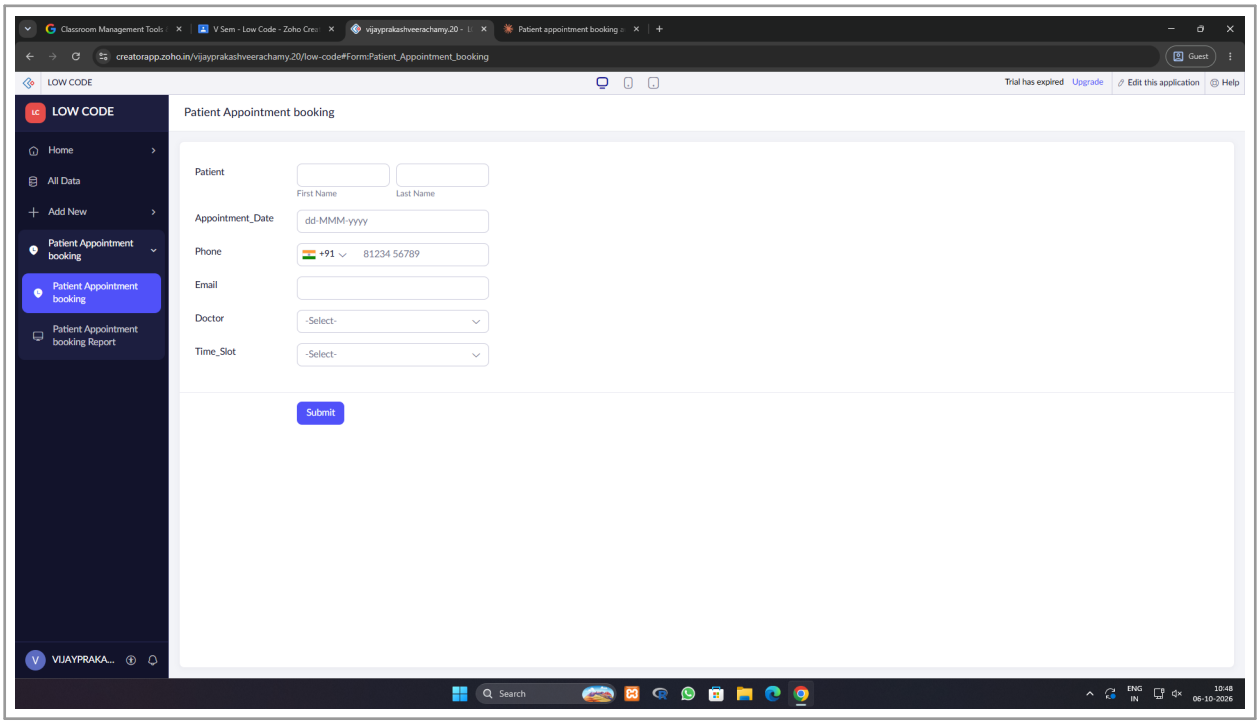


Figure 1: Live Patient Appointment Booking Form

3.2 Form Workflows — Validations on Form Submission

The Form Workflows screen lists the Validate appointment workflow under the Created event, with its status showing Enabled, confirming the validation rule is active on the form.

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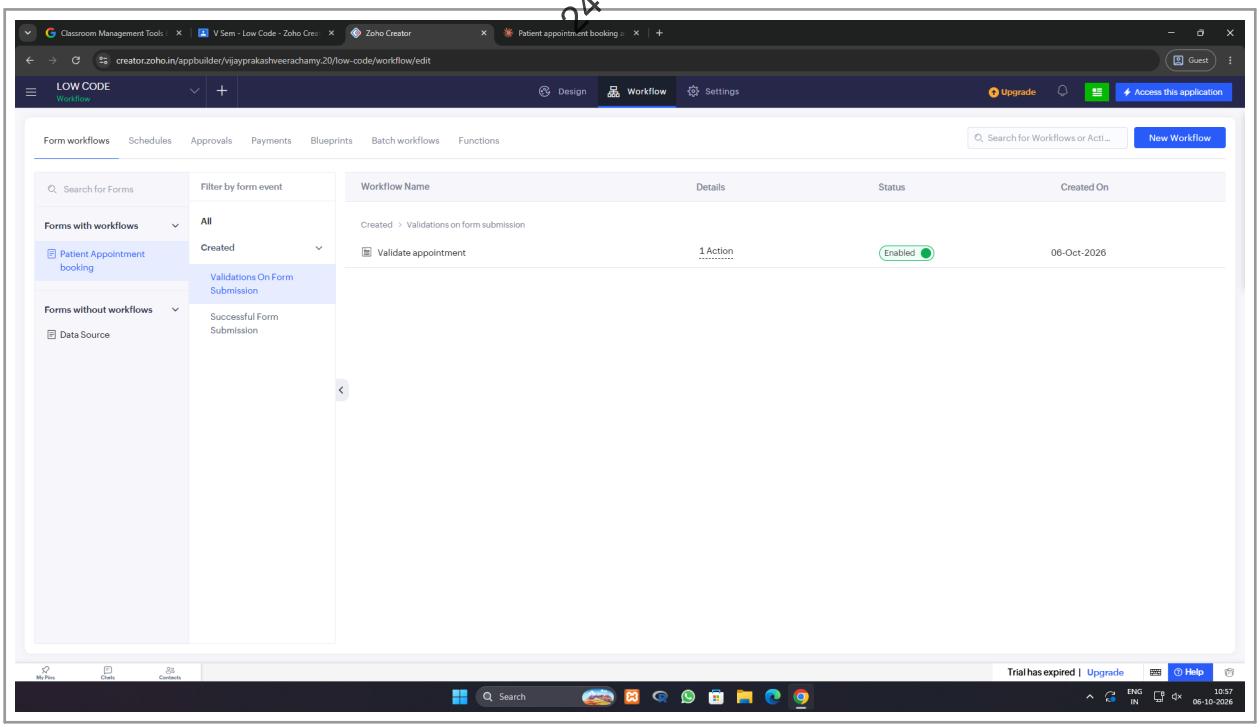


Figure 2: Form Workflows List – "Validate appointment" Enabled

3.3 New Workflow Configuration Screen

This is the setup screen used to create a workflow: the Patient Appointment booking form is chosen, the record event and form event are selected, and the workflow is named before it is created.

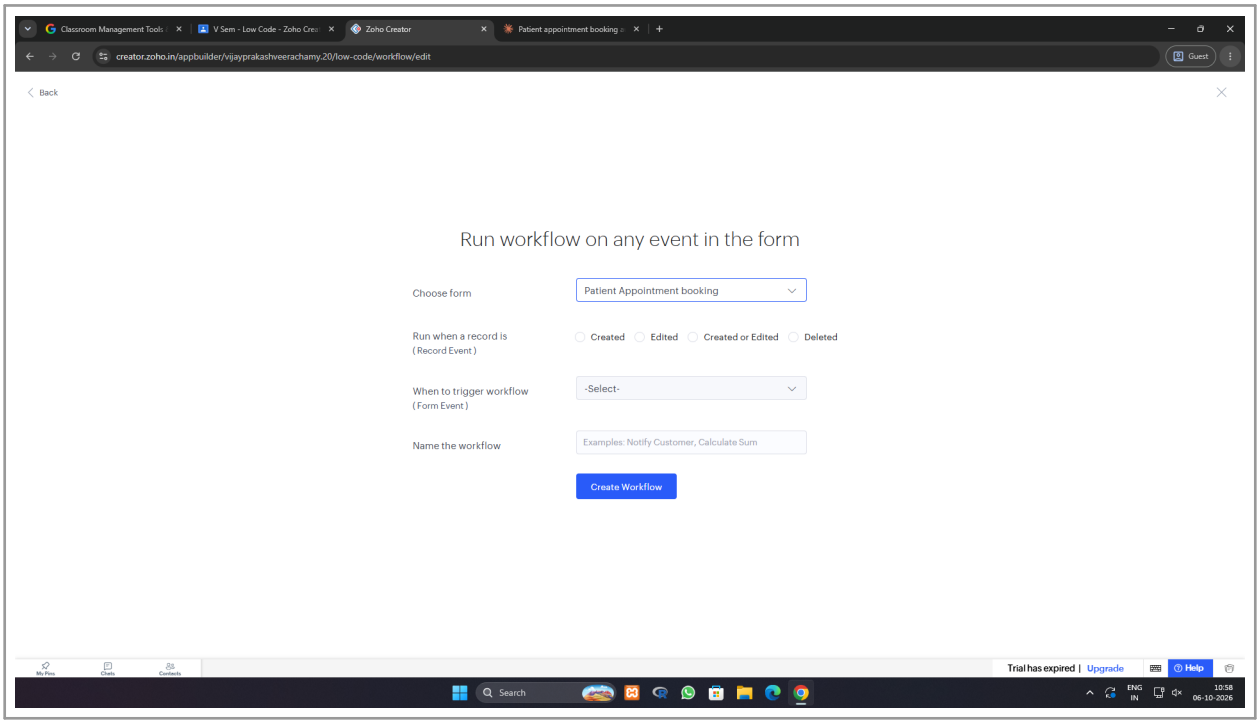


Figure 3: New Workflow Setup – Choosing the Form and Trigger Event

3.4 Workflow Builder — Validate Appointment Action

Inside the workflow builder, the Validate appointment workflow is set to Execute without condition and runs a Deluge script action, shown here in the inline code panel.

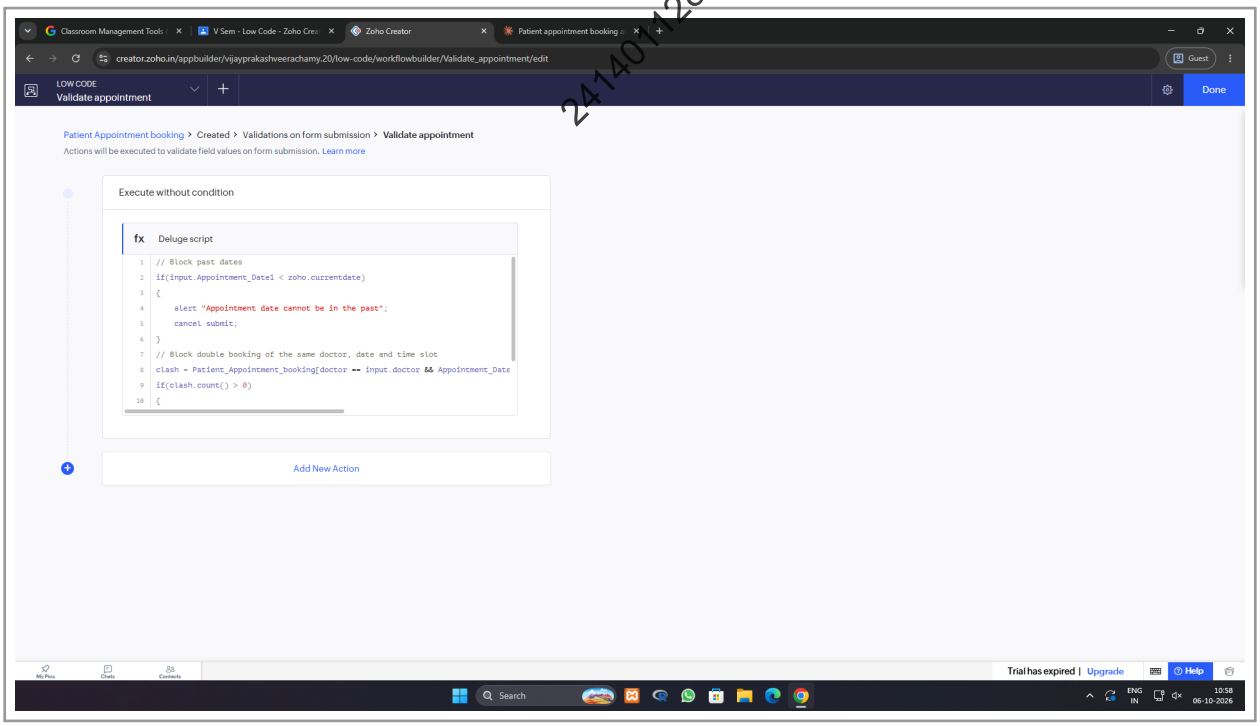


Figure 4: Workflow Builder – Deluge Script Action for "Validate appointment"

3.5 Deluge Script — Validate Appointment (full view)

The full Deluge editor shows the two validation blocks in detail: the past-date check, which cancels the submission with an alert if the appointment date has already passed, and the clash check, which cancels the submission if the same doctor, date and time slot are already booked.

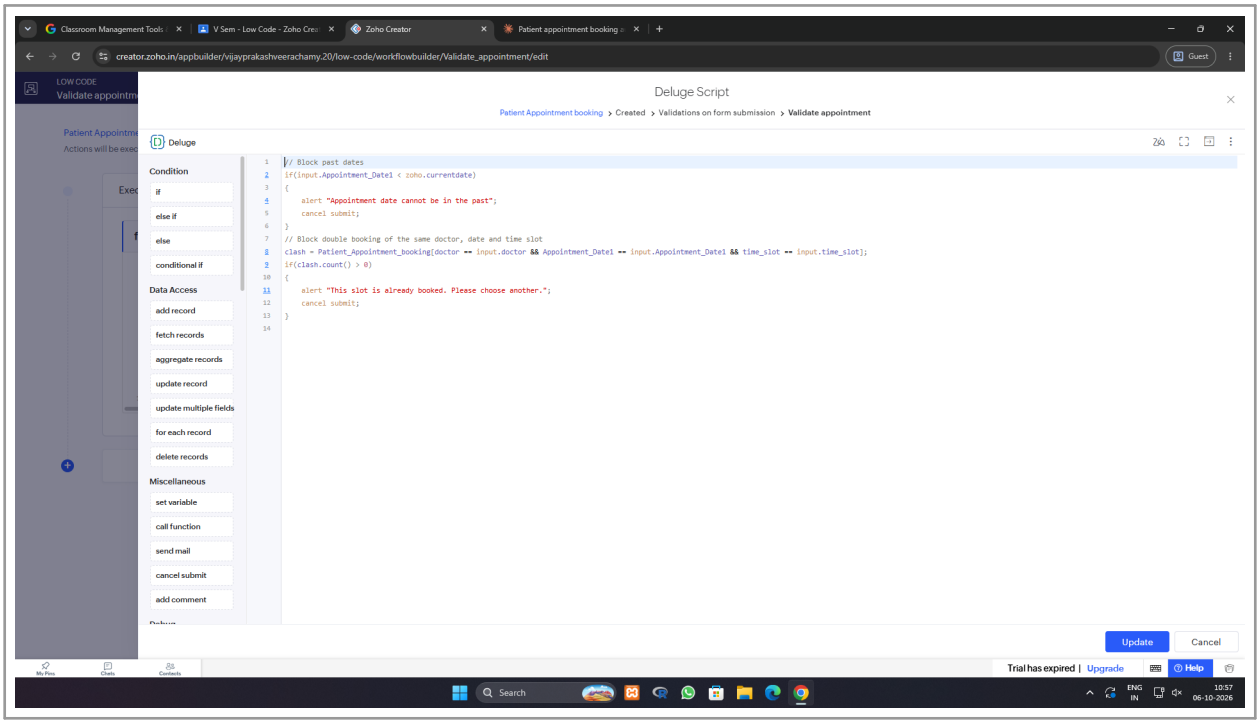


Figure 5: Deluge Script – "Validate appointment" (Full Editor View)

3.6 Deluge Script — Send Confirmation

The Send Confirmation workflow (under Successful form submission) runs a sendmail action, dynamically composing the subject and message from the submitted record's patient name, doctor, appointment date and time slot, and sending it to the patient's Email address.

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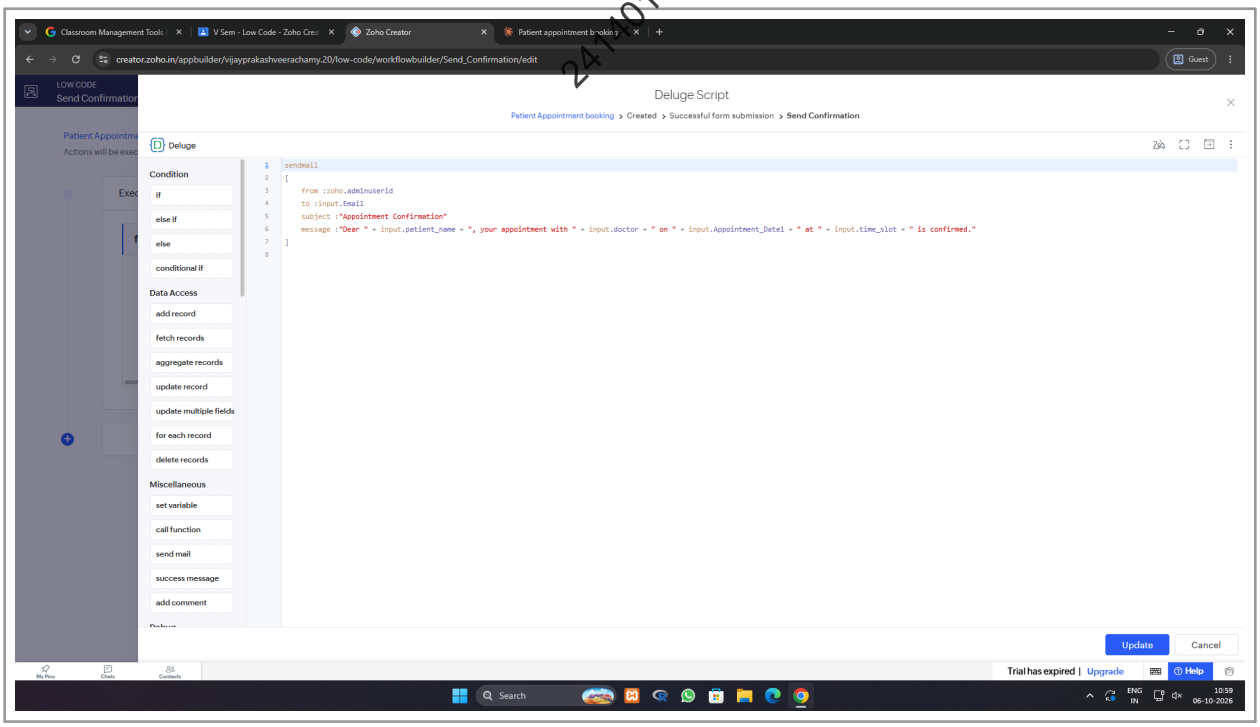


Figure 6: Deluge Script – "Send Confirmation" (sendmail Action)

3.7 Debugging Output — Confirmation Email Received

After submitting a test booking, the recipient's inbox received the automatically generated confirmation mail with the patient name, doctor and time slot correctly merged into the message, confirming that both the validation workflow and the sendmail workflow executed successfully end-to-end.

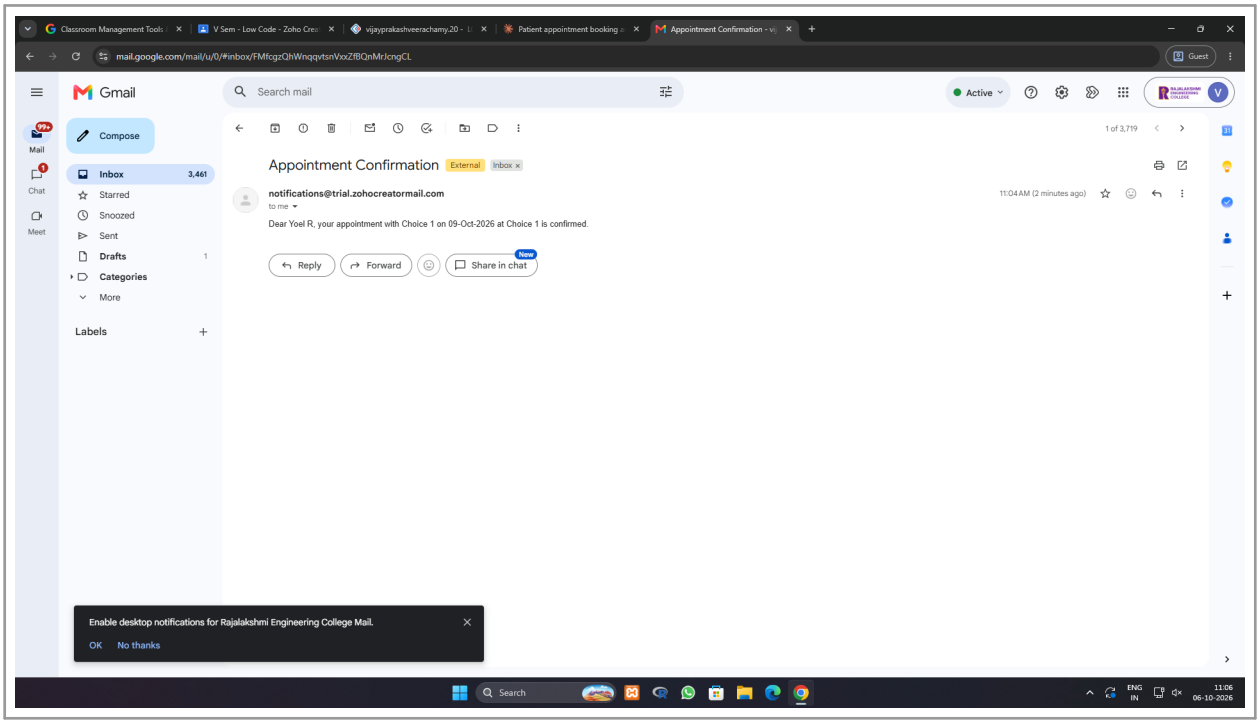


Figure 7: Received "Appointment Confirmation" Email in Gmail

4. Result

The business logic behind the Patient Appointment Booking Form was successfully automated using Deluge scripting and Zoho Creator's Workflow engine. The Validate appointment workflow blocks both past-dated and clashing bookings at the point of submission, while the Send Confirmation workflow automatically emails the patient once a booking passes validation.

Both workflows were tested against live form submissions and debugged by tracing the Deluge script logic alongside the actual email received, confirming that the automation runs correctly from form submission through to patient notification, without any manual intervention from the front desk.

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